

RESEARCH INTERESTS

Autonomous and multi-agent systems, with a focus on exploring how structured representations and constraints shape agent behavior, coordination, and failure modes, as observed through prompt-level schemas and interactive agent prototypes in complex tasks. I am particularly interested in bridging agent architectures with human–AI interaction, studying how procedural brittleness emerges in agent-driven systems and how interactive and computational media can be used to make agent behavior more interpretable, controllable, and trustworthy. My interests also include scalable, safety-oriented AI system design, especially in settings involving long-term adaptation and sustained agent–user interaction.

RESEARCH / TECHNICAL PROJECTS

Emotion Landscape — Interactive Storytelling System New York City, USA
Columbia DSL (Digital Storytelling Lab) 2025 - Present

- Investigated how structured prompting strategies affect emergent behavior in interactive, agent-driven narrative environments. Designed modular interaction logic to study coordination between narrative agents and user actions.
- Built an assets pipeline and explored affective design patterns for interactive spaces.

Prompt Markdown+
Independent Research 2025 - Present

- Designed a structured prompt language to improve controllability and interpretability, and to study how abstraction layers affect agent failure modes and user mental models in human–AI interaction.
- Built an early prototype and evaluation plan to measure agent consistency under constrained prompt schemas (preparing user study).

Interactive Engine (In.js)
Independent Research 2025 - Present

- Built a modular JavaScript-based interactive engine to experiment with agent-driven narrative logic.
- Exploring how architectural choices affect scalability, modularity, and agent coordination in interactive systems.

TECHNICAL SKILLS

- Programming: Python, Java, C/C++, JavaScript
- Systems/Dev: Linux, Git, CLI
- AI/Agents: Prompt Engineering, Multi-agent Prompting, HCI Prototyping
- Prototyping & Media: Interactive demos, Video-based documentation, Godot

RESEARCH TRAINING & TECHNICAL EXPERIENCE

Digital Storytelling Lab (DSL), Columbia University 2025 - Present

- Participated in project-based research on interactive systems and computational storytelling, with an emphasis on iterative prototyping and analysis of agent behavior, user interaction, and narrative dynamics.
- Selected demos: [YouTube link](#)

Columbia DevFest Hackathon 2025.11
Participant

- Participated in rapid prototyping of interactive AI systems; responsibilities included prompt engineering, narrative design, and interface logic.

ACM-ICPC 2025 2025.11
Volunteer

- Assisted technical operations; gained exposure to competitive programming workflows and algorithmic problem-solving culture.

NSF(The National Science Foundation) I-Corps
Propelus Certificate of Completion

Princeton University
Spring 2024

- Completed National Science Foundation (NSF) Innovation Corps (I-Corps™) Northeast Hub -funded training in research commercialization, customer discovery, and early-stage product validation.
- Received micro-funding to identify initial customers and define a prototype concept, collaborating with entrepreneurs and mentors across NSF's innovation network.
- Designed an AI system informed by Buddhist contemplative principles to support reflective user interaction. Secured NSF I-Corps funding and conducted over 100 user interviews to validate needs among monasteries, practitioners, and scholars.

EDUCATION

Columbia University
B.A. in Computer Science & Cognitive Science

New York City, USA
2025 - Present

- Dean's List, GPA 3.93 / 4.0

Fall 2025

Chinese Academy of Social Sciences (CASS)
M.Phil., Humanities (textual and visual analysis; narrative structures)

Beijing, China

- Overall GPA: 3.70/4.0

Beijing University of Technology (BJUT)
Bachelor of Engineering, Civil Engineering

Beijing, China